

Perfect Exact Pairs in Coarse-Description Classes

An attack on the least-Turing-representative question

Prepared for Vladimir Reshetnikov

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Abstract

We investigate the least-representative question selected as C1 in the supplied Turing-degree survey. For a set X , let $\mathcal{C}_X$ be the sets differing from X on a density-zero set, and let $\mathcal{I}_X$ be the sets computable from every member of $\mathcal{C}_X$. We give a conventional proof that $\mathcal{C}_X$ contains a perfect family of pairwise Turing-incomparable members such that the common lower cone of any two is exactly $\mathcal{I}_X$. The construction can simultaneously control intersections with the lower cones of any prescribed countable family of oracles. Its main ingredient is a local recovery lemma in a complete metric on $\mathcal{C}_X$: information computed on a dense open-region subset is already computable from every coarse description.

An explicit computably enumerable, many-one complete set X has $\mathcal{I}_X = \mathbf{Comp}$ but no computable coarse description. Consequently its uniform and nonuniform coarse-equivalence classes contain Turing minimal pairs, have no least Turing degree, and have no countable coinital family of Turing degrees. Relativized examples have any prescribed principal core. All infinite arguments are written out; the accompanying program checks only finite arithmetic lemmas.

Status and attribution. The negative answer to the literal C1 question already follows from results published in 2016; it is not claimed here as a new resolution of an open problem. The perfect-family and simultaneous exact-pair development is the stronger result pursued in this note. Its proof is supplied in full, but its novelty has not been established, and it has not been independently refereed or checked by a proof-assistant kernel.

Contents

1	Target, result, and correction of the research map	3
2	Descriptions, reducibility, and the common-information ideal	4
3	A complete metric on one coarse-description class	5
4	Local recovery and simultaneous cone avoidance	6
5	A perfect family of exact pairs	7
6	Computable approximability forces a small core	9
7	An explicit complete c.e. counterexample	10
8	Minimal pairs, no least degree, and uncountable coinitality	12
9	Which nonuniform coarse classes have least representatives?	13
10	Relativized examples with any principal core	14
11	Proof audit, limitations, and continuation targets	15
A	Finite checks and reproduction	16
	References	17

1 Target, result, and correction of the research map

The uploaded report [1], entry C1, asks whether every nonuniform coarse-equivalence class contains a representative of least Turing degree. For total functions, its statement is

$$\forall f \in \mathbb{N}^{\mathbb{N}} \exists g \equiv_{\text{nc}} f \forall h \equiv_{\text{nc}} f \quad g \leq_T h. \quad (1)$$

The report proposes a concrete negative route: find a nonzero coarse class containing two representatives that form a Turing minimal pair. This is the route taken here, with an additional perfect-set strengthening.

1.1 What the source check changes

Gerdes’s 2025 preprint explicitly asks the least-degree question in Question 7 [2]. However, the invariant used below is already Definition 3.1 of Hirschfeldt–Jockusch–Kuyper–Schupp [3], where it is written X^c . Their Theorem 4.3, attributed to Igusa, gives a trivial invariant when the coarse computability bound is 1, including examples that are not coarsely computable. Those two properties alone negate (1), as the next paragraph proves. The journal publication is from 2016. Thus the literal C1 entry is not a defensible novelty target, despite its appearance in the later question list.

Suppose every coarse description of X has only computable information in common, while X has no computable coarse description. Any least representative g of the coarse class would be computable from every coarse description of X , since each such description itself belongs to the class. The graph of g would therefore be computable. But a computable member of the class would provide a computable coarse description of X , a contradiction. This argument works for both uniform and nonuniform coarse equivalence.

The status correction does not end the investigation. The older invariant statement does not itself present the minimal-pair certificate requested by the survey, nor the exact-pair and perfect-family results proved here. The remainder develops those stronger statements without relying on randomness, a priority minimal-pair theorem, or an imported cone-avoidance theorem.

1.2 The principal structural theorem

Write $\downarrow_T B = \{A \subseteq \mathbb{N} : A \leq_T B\}$. The word *exact* refers to equality of common lower cones with an entire ideal, not merely to the exclusion of one selected noncomputable set.

Theorem 1.1 (Perfect exact pairs; proved in Section 5). *For every $X \subseteq \mathbb{N}$, put*

$$\mathcal{C}_X = \{D \subseteq \mathbb{N} : \rho_n(D \triangle X) \longrightarrow 0\}, \quad \mathcal{I}_X = \bigcap_{D \in \mathcal{C}_X} \downarrow_T D.$$

There is a perfect set $P \subseteq \mathcal{C}_X$ of pairwise Turing-incomparable members such that

$$D, E \in P, D \neq E \implies (\downarrow_T D) \cap (\downarrow_T E) = \mathcal{I}_X. \quad (2)$$

Moreover, given any countable family of oracles $(B_j)_{j \in \mathbb{N}}$, P may be chosen so that, for every $D \in P$ and every j ,

$$(\downarrow_T D) \cap (\downarrow_T B_j) = \mathcal{I}_X \cap (\downarrow_T B_j). \quad (3)$$

No effective presentation of (B_j) is assumed or needed.

Taking $B_0 = X$ gives an exact partner for the original, prescribed X , not merely two newly chosen descriptions. For the explicit X constructed in Section 7, $X \equiv_T \emptyset'$ and $\mathcal{I}_X = \text{Comp}$. Thus every member of the selected perfect family forms a Turing minimal pair with $\emptyset'$, as well as with every other member of that family.

1.3 What is and is not being claimed

The negative answer to C1 is a consequence of established work, not a new breakthrough. The mathematical contribution attempted here is the fully proved perfect exact-pair formulation and its applications. A targeted literature search did not establish whether these stronger formulations have already appeared, perhaps as consequences of generic-description arguments. Their novelty therefore remains unverified.

The effective-dense problem C2 is not answered. Nor do the results prove that the exhibited coarse classes contain no *minimal* Turing degrees. A least degree is below every representative; a minimal degree need only have no strictly smaller representative. These are different assertions.

2 Descriptions, reducibility, and the common-information ideal

Identify a set with its characteristic function. For $n \geq 1$, define

$$\rho_n(S) = \frac{|S \cap [0, n)|}{n}, \quad \delta(Y, Z) = \limsup_{n \rightarrow \infty} \rho_n(Y \triangle Z).$$

Thus $D \in \mathcal{C}_X$ exactly when $\delta(D, X) = 0$. Because prefix discrepancies are nonnegative, a zero limsup is equivalent to convergence to zero.

For total functions f, g , define $f \leq_{\text{nc}} g$ to mean that every coarse description of g computes some coarse description of f . In the uniform version $f \leq_{\text{uc}} g$, one Turing functional must work for every source description. These are the conventions of the supplied report and Gerdes [2]. We write $[X]_r$ for the equivalence class under r , with $r \in \{\text{nc}, \text{uc}\}$.

Binary source sets cause no ambiguity between binary and natural-number-valued descriptions. Replace every nonbinary answer in a description of X by zero. This transformation is computable and does not create a disagreement at any previously correct position. Consequently the binary descriptions suffice for the common-information questions below. Total functions are treated as oracles by coding their graphs as sets; this coding preserves Turing degree.

Lemma 2.1 (Equality of description classes). *If $D \in \mathcal{C}_X$, then $\mathcal{C}_D = \mathcal{C}_X$. In particular $D \equiv_{\text{uc}} X$, and hence $D \equiv_{\text{nc}} X$.*

Proof. For each prefix, disagreement with X is contained in the union of disagreement with D and $D \triangle X$. Taking upper densities gives

$$\delta(E, X) \leq \delta(E, D) + \delta(D, X).$$

The same inequality with X, D exchanged proves the equivalence of the two zero-discrepancy conditions. The identity functional is a coarse reduction in both directions. The argument also applies to natural-number-valued coarse descriptions. $\square$

Definition 2.2 (Common-information ideal). For $X \subseteq \mathbb{N}$, define

$$\mathcal{I}_X = \{A \subseteq \mathbb{N} : (\forall D \in \mathcal{C}_X) A \leq_{\text{T}} D\}.$$

We call this the *core* only as an abbreviation for the displayed invariant; the invariant itself is established terminology and notation X^c in [3, Definition 3.1].

Lemma 2.3. *The family $\mathcal{I}_X$ is a countable Turing ideal: it contains every computable set, is downward closed under $\leq_{\text{T}}$, and is closed under finite Turing joins. It is contained in $\downarrow_{\text{T}} X$. If $X \leq_{\text{nc}} Y$, then $\mathcal{I}_X \subseteq \mathcal{I}_Y$. In particular the core is invariant under both coarse equivalences.*

Proof. Every oracle computes every computable set. If $A \leq_T B$ and every description computes B , every description computes A . If every description computes both A and B , it computes their join. Since X itself is a description of X , each member of the core is computable from X . There are only countably many oracle programs, so $\downarrow_T X$ and hence $\mathcal{I}_X$ are countable.

For monotonicity, fix $A \in \mathcal{I}_X$ and $D \in \mathcal{C}_Y$. The assumed reduction provides an $E \in \mathcal{C}_X$ with $E \leq_T D$. Then $A \leq_T E \leq_T D$. Since this holds for every D , $A \in \mathcal{I}_Y$. Applying this in both directions proves invariance. Uniform reducibility implies nonuniform reducibility. $\square$

Remark 2.4 (Two different equivalence classes). The space $\mathcal{C}_X$ is the equivalence class of X modulo density-zero symmetric difference. It is a subset, generally a proper subset, of the coarse-reducibility class $[X]_r$. We construct witnesses in the smaller space. Confusing these two classes would invalidate the least-representative argument.

3 A complete metric on one coarse-description class

The asymptotic discrepancy δ is identically zero on $\mathcal{C}_X \times \mathcal{C}_X$, so it cannot provide a useful topology there. Instead use every prefix:

$$d(Y, Z) = \sup_{n \geq 1} \rho_n(Y \triangle Z). \quad (4)$$

This is a metric on all of $2^{\mathbb{N}}$. We use its restriction to $\mathcal{C}_X$; this restriction is not the topology inherited from ordinary Cantor space.

For a finite binary string σ and a set Z , write $\sigma \star Z$ for Z with its first $|\sigma|$ bits replaced by σ .

Lemma 3.1 (Finite-prefix facts). *The following statements hold.*

- (i) *If $Y(j) \neq Z(j)$, then $d(Y, Z) \geq 1/(j+1)$. Hence each ordinary finite cylinder is open in the d topology.*
- (ii) *If $L = |\sigma| > 0$, then*

$$d(\sigma \star Z, Z) = \max_{1 \leq n \leq L} \frac{|\{j < n : \sigma(j) \neq Z(j)\}|}{n}. \quad (5)$$

For the empty string this distance is zero.

- (iii) *The triangle inequality holds for d and for δ .*

Proof. A disagreement at j contributes at least one error to the prefix of length $j+1$. In particular a d -ball of radius less than $1/L$ fixes the first L bits. For (ii), after position L the number of disagreements is constant, so its proportion cannot exceed the proportion at L . For (iii), use the inclusion $Y \triangle W \subseteq (Y \triangle Z) \cup (Z \triangle W)$ at each prefix, and take a supremum or a limsup. Symmetry is immediate, and (i) supplies separation, completing the metric verification. $\square$

Proposition 3.2 (Polish description space). *The metric space $(\mathcal{C}_X, d)$ is nonempty, complete, separable, and has no isolated points. For every $Z \in \mathcal{C}_X$, the finite modifications of Z are dense in $(\mathcal{C}_X, d)$.*

Proof. Nonemptiness follows from $X \in \mathcal{C}_X$. Let (D_k) be d -Cauchy. For each coordinate j , the first part of Lemma 3.1 shows that $D_k(j)$ is eventually constant. Let $D(j)$ be this eventual value.

Given $\varepsilon > 0$, choose K so that $d(D_k, D_l) < \varepsilon/2$ whenever $k, l \geq K$. Fix $k \geq K$ and a prefix length n . For sufficiently large l , $D_l \upharpoonright n = D \upharpoonright n$, whence $\rho_n(D_k \triangle D) \leq \varepsilon/2$. This holds for every n , so $d(D_k, D) \leq \varepsilon/2 < \varepsilon$. Thus $D_k \rightarrow D$ in d . Also

$$\delta(D, X) \leq d(D, D_k) + \delta(D_k, X) = d(D, D_k),$$

which tends to zero. Therefore $D \in \mathcal{C}_X$, proving completeness.

Fix $Y, Z \in \mathcal{C}_X$. Put $Z_N = (Y \upharpoonright N) \star Z$, a finite modification of Z . The first N bits of Z_N and Y agree, and for $n > N$ their errors are a subset of $Y \triangle Z$ below n . Therefore

$$d(Z_N, Y) \leq \sup_{n > N} \rho_n(Y \triangle Z) \longrightarrow 0. \quad (6)$$

The convergence holds because $\delta(Y, Z) = 0$. Thus the countable set of finite modifications of Z is dense, proving separability as well.

Finally, flip just bit j of an arbitrary $Y \in \mathcal{C}_X$. The result remains in $\mathcal{C}_X$ and has distance exactly $1/(j+1)$ from Y . Taking j arbitrarily large proves that no point is isolated. $\square$

A set is *nowhere dense* if its closure has empty interior, and *meager* if it is a countable union of nowhere dense sets. We use the Baire category theorem in the form: a nonempty complete metric space cannot be covered by countably many nowhere dense sets. Its usual nested-ball proof also shows that the complement of such a union is dense. These topological statements are used with d , not with δ .

4 Local recovery and simultaneous cone avoidance

Fix an enumeration (Φ_e) of oracle Turing functionals. For a set A , write

$$S_{e,A} = \{D \in \mathcal{C}_X : \Phi_e^D = A\}.$$

Equality here includes totality on every input and binary output. A computation $\Phi_e^\sigma(n) \downarrow = v$ uses only oracle bits within the finite string σ .

Lemma 4.1 (Local recovery). *If $S_{e,A}$ is dense in a nonempty d -open set $U \subseteq \mathcal{C}_X$, then $A \in \mathcal{I}_X$.*

Proof. Fix any $Z \in \mathcal{C}_X$. Finite modifications of Z are dense, so choose a finite modification Z^* in U . Choose a positive rational q with $B_d(Z^*, q) \subseteq U$, where the ball is taken within $\mathcal{C}_X$. The finite modification and q may depend on Z ; they will be constants in the reduction below.

To compute $A(n)$ using Z^* , dovetail over finite strings σ and finite simulations of $\Phi_e^\sigma(n)$. Accept a convergent simulation with value v only if

$$\frac{|\{j < k : \sigma(j) \neq Z^*(j)\}|}{k} < q \quad (1 \leq k \leq |\sigma|). \quad (7)$$

Each test is a finite, Z^* -decidable calculation. No test of membership in $\mathcal{C}_X$, totality, or asymptotic density is performed.

Termination. Density gives a $D \in S_{e,A} \cap B_d(Z^*, q)$. A finite initial segment $\sigma \prec D$ witnesses convergence on input n . Every prefix discrepancy of D from Z^* is at most $d(D, Z^*) < q$, so this σ passes (7). The dovetailing eventually finds an accepted computation.

Correctness. For any accepted σ , the set $W = \sigma \star Z^*$ belongs to $\mathcal{C}_X$, because it is a finite modification of Z^* . By (5) and the finitely many strict inequalities in (7), $d(W, Z^*) < q$. Thus $W \in U \cap [\sigma]$. The intersection is a nonempty d -open set. Density of $S_{e,A}$ in U provides $D' \in S_{e,A} \cap U \cap [\sigma]$. The same finite computation runs with oracle D' , so the accepted value is $v = A(n)$.

The search computes A from Z^* . Since Z^* is a fixed finite modification of Z , $Z^* \equiv_T Z$, and therefore $A \leq_T Z$. The choice of Z was arbitrary, which proves $A \in \mathcal{I}_X$. $\square$

Remark 4.2 (The nonuniformity is essential). The proof asserts $\forall Z \in \mathcal{C}_X \exists i (\Phi_i^Z = A)$. It does not supply one index working for every Z , an effective choice of the finite patch, or an effective choice of the rational radius. Hard-coding finitely many constants proves an ordinary Turing reduction; finding those constants uniformly would be a stronger assertion.

Theorem 4.3 (All-or-meager cone dichotomy). *For any $A \subseteq \mathbb{N}$, the cone $\{D \in \mathcal{C}_X : A \leq_T D\}$ is either all of $\mathcal{C}_X$ or a meager subset of it. More precisely, if $A \notin \mathcal{I}_X$, each $S_{e,A}$ is nowhere dense.*

Proof. When $A \in \mathcal{I}_X$, the cone is all of the space by definition. Otherwise, if $\overline{S_{e,A}}$ had nonempty interior U , $S_{e,A}$ would be dense in U . Lemma 4.1 would imply $A \in \mathcal{I}_X$, a contradiction. Thus each $S_{e,A}$ is nowhere dense. The cone is their countable union. $\square$

Corollary 4.4 (Countable avoidance). *Given a countable family (A_j) with every $A_j \notin \mathcal{I}_X$, the set*

$$\{D \in \mathcal{C}_X : (\forall j) A_j \not\leq_T D\}$$

is comeager, and in particular is nonempty and dense. No effective enumeration of the family is required.

Proof. Remove all sets S_{e,A_j} . This removes a countable union of nowhere dense sets. Proposition 3.2 and the Baire category theorem apply. $\square$

Theorem 4.5 (Exact partners for prescribed oracles). *For every countable family (B_j) there is $D \in \mathcal{C}_X$ such that*

$$(\downarrow_T D) \cap (\downarrow_T B_j) = \mathcal{I}_X \cap (\downarrow_T B_j) \quad \text{for all } j.$$

The set of suitable D is comeager. In particular, some $D \in \mathcal{C}_X$ satisfies

$$(\downarrow_T D) \cap (\downarrow_T X) = \mathcal{I}_X. \tag{8}$$

Proof. There are countably many sets computable from any one oracle B_j , and hence countably many in

$$\mathcal{A} = \bigcup_j ((\downarrow_T B_j) \setminus \mathcal{I}_X).$$

Avoid all members of $\mathcal{A}$ by Corollary 4.4. If A is computable from both the resulting D and B_j , it cannot belong to $(\downarrow_T B_j) \setminus \mathcal{I}_X$, and thus belongs to the right-hand side. Conversely every member of $\mathcal{I}_X$ is computable from D , which proves the reverse inclusion. Take $B_0 = X$ and use $\mathcal{I}_X \subseteq \downarrow_T X$ for (8). $\square$

This avoids an important logical shortcut. The assertion that each individual non-core set can be avoided does not, by itself, give one oracle avoiding all of them simultaneously. The nowhere-density proof and Baire category are the additional ingredients that make simultaneous avoidance valid.

5 A perfect family of exact pairs

We now prove Theorem 1.1, including the word *perfect*. We give the finite-level fusion directly, rather than invoking a perfect-set selection theorem as an unexplained black box.

Lemma 5.1 (Bad pair relations are nowhere dense). *For $e, f \in \mathbb{N}$, put*

$$R_{e,f} = \{(D, E) \in \mathcal{C}_X^2 : \Phi_e^D = \Phi_f^E = A \in 2^{\mathbb{N}} \text{ for some } A \notin \mathcal{I}_X\}.$$

Then $R_{e,f}$ is nowhere dense in the product of the d topologies.

Proof. Suppose to the contrary that $R_{e,f}$ is dense in a nonempty open rectangle $U \times V$. Such a rectangle would exist if its closure had interior. Choose $(D_0, E_0) \in R_{e,f} \cap (U \times V)$, and let $A = \Phi_e^{D_0} = \Phi_f^{E_0} \notin \mathcal{I}_X$.

Fix n . We claim that every convergent computation $\Phi_e^D(n)$ with $D \in U$ has value $A(n)$. Otherwise there would be a $D \in U$ whose computation converges to a different value v . Choose finite initial segments witnessing this computation and $\Phi_f^{E_0}(n) = A(n)$. Intersect their cylinders with U and V . The resulting nonempty open rectangle consists of pairs with unequal outputs on input n , and is therefore disjoint from $R_{e,f}$. This contradicts the assumed density. The claim holds for every n .

The projection of $R_{e,f} \cap (U \times V)$ onto U is dense in U : every nonempty open subset of U , multiplied by V , meets $R_{e,f}$. Every oracle in this projection makes Φ_e total. By the preceding claim its output is exactly A . Therefore $S_{e,A}$ is dense in U . Lemma 4.1 gives $A \in \mathcal{I}_X$, contradicting its choice. $\square$

Theorem 5.2 (Perfect exact-pair theorem). *There exists a continuous injection $F : 2^{\mathbb{N}} \rightarrow (\mathcal{C}_X, d)$ whose image P satisfies (2). It can be chosen to satisfy (3) for any prescribed countable family (B_j) , and so that $P \cap \mathcal{I}_X = \emptyset$. Its image is perfect in the ordinary Cantor topology, and its members are pairwise Turing incomparable.*

Proof. For each (e, f) let

$$O_{e,f} = \mathcal{C}_X^2 \setminus \overline{R_{e,f}}.$$

By Lemma 5.1, these are open dense sets. They are indexed by *ordered* pairs of functionals; there is no symmetry assumption.

We also need countably many unary open dense requirements. For each set A computable from some B_j but not in $\mathcal{I}_X$, and each e , include $\mathcal{C}_X \setminus \overline{S_{e,A}}$. Include as well $\mathcal{C}_X \setminus \{A\}$ for every $A \in \mathcal{I}_X \cap \mathcal{C}_X$. These last sets are open dense because singletons are closed and no point is isolated. There are only countably many requirements, by Lemma 2.3. Enumerate them as (V_i) , repeating the whole space if necessary. This enumeration is an ordinary set-theoretic enumeration, not a computable one.

Construct, for each finite binary string s , a nonempty open metric ball $U_s \subseteq \mathcal{C}_X$. Write $K_s = \overline{U_s}$, with closure within $\mathcal{C}_X$. Require

$$K_{s0} \cup K_{s1} \subseteq U_s, \quad K_{s0} \cap K_{s1} = \emptyset, \quad (9)$$

$$\text{diam}(K_s) \leq 2^{-|s|} \quad \text{if } |s| > 0, \quad (10)$$

$$K_s \subseteq V_i \quad \text{if } i \leq m = |s|, \quad m \geq 1, \quad (11)$$

$$K_s \times K_t \subseteq O_{e,f} \quad \text{if } s \neq t, \quad |s| = |t| = m \geq 1, \quad e, f \leq m. \quad (12)$$

Start with any nonempty open ball for the root; impose no diameter condition at level zero.

Here is the complete extension step. In each current parent ball choose two distinct points. Small enough balls around them have disjoint closures contained in the parent and meet the next diameter bound. Such points exist because the space has no isolated points. This gives preliminary child balls at the next level.

There are finitely many children and finitely many unary requirements with $i \leq m$ at level m . Intersect each child successively with those open dense sets, choosing a smaller nonempty ball whose closure lies in the intersection. Then list all finitely many ordered distinct child pairs (s, t) and all $e, f \leq m$. For the next pair requirement, the product of the current two balls is a nonempty open rectangle, so it meets $O_{e,f}$. Because $O_{e,f}$ is open, choose smaller balls in the two coordinates whose product of closures is contained in $O_{e,f}$. Shrink just those two child balls. Every previously established requirement persists under shrinking. After finitely many steps all requirements at that level hold. This completes the recursive construction of the balls.

For $p \in 2^{\mathbb{N}}$, the sets $(K_{p \upharpoonright m})_{m \geq 1}$ are nonempty closed, nested, and have diameters tending to zero. To see that they have a unique common point without any compactness assumption, select one point from each. The nesting and diameter bounds make this sequence Cauchy. Completeness puts its limit in every closed set, and the vanishing diameters give uniqueness. Define $F(p)$ to be that point. Shared long prefixes make the images close, so F is continuous. Disjoint sibling closures make it injective.

Let $p \neq q$, and fix (e, f) . At a sufficiently large level m , the two prefixes are distinct and $m \geq e, f$. Requirement (12) gives $(F(p), F(q)) \in O_{e,f}$, and thus this pair is not in $R_{e,f}$. If a set A is computable from both images, some such pair of functionals computes it. The excluded relation therefore implies $A \in \mathcal{I}_X$. The reverse inclusion holds because every image is in $\mathcal{C}_X$. This proves (2).

The unary requirements exclude every $S_{e,A}$ for $A \leq_T B_j$ outside the core. The argument of Theorem 4.5 then proves (3). The singleton requirements ensure $P \cap \mathcal{I}_X = \emptyset$. If distinct $D, E \in P$ satisfied $D \leq_T E$, the set D itself would lie in their common lower cone, hence in $\mathcal{I}_X$, contradicting this exclusion. This proves pairwise incomparability.

Finally, the identity map from $(\mathcal{C}_X, d)$ into ordinary Cantor space is continuous by Lemma 3.1. Composing it with F yields a continuous injection from compact Cantor space into a Hausdorff space. It is therefore a homeomorphism onto its image. The image is compact, hence closed, and has no isolated points. Thus P is perfect in the usual sense as well. $\square$

Remark 5.3 (A useful safeguard). A perfect family of Turing minimal pairs does not alone disprove a least-degree claim about its surrounding coarse class. The computable coarse class also contains many noncomputable representatives. The later application separately proves that its coarse class contains no computable representative.

6 Computable approximability forces a small core

We next produce a core to which the exact-pair theorem can be applied. The following proof is independent of the older cone-avoiding compactness proof. Its unrelativized conclusion is the result attributed to Igusa in [3, Theorem 4.3]; no novelty is claimed for that conclusion.

Definition 6.1. For an oracle Z , let $\text{AP}_Z(X)$ denote the property

$$\forall \varepsilon > 0 \exists C \leq_T Z \quad \delta(C, X) < \varepsilon. \quad (13)$$

The sequence of approximating sets need not be uniformly computable in Z . For the computable oracle, this is the usual assertion that the coarse computability bound is 1.

Lemma 6.2 (Patching an approximate center). *If $\delta(C, Y) < \eta$, there is a finite modification C_0 of C such that $d(C_0, Y) < \eta$.*

Proof. Choose η' strictly between $\delta(C, Y)$ and η . For some N , all prefixes of length $n > N$ have discrepancy less than η' . Put $C_0 = (Y \upharpoonright N) \star C$. Prefixes of length at most N have no errors, and longer prefixes have no more errors than $C \Delta Y$. Consequently $d(C_0, Y) \leq \eta' < \eta$. The finite patch does not change the Turing degree. $\square$

Theorem 6.3 (Relative approximability bound). *If $\text{AP}_Z(X)$, then $\mathcal{I}_X \subseteq \downarrow_T Z$. In particular, $\text{AP}_\emptyset(X)$ implies $\mathcal{I}_X = \text{Comp}$.*

Proof. Fix $A \in \mathcal{I}_X$. Every member of $\mathcal{C}_X$ computes A , so $\mathcal{C}_X = \bigcup_e S_{e,A}$. By the Baire category theorem, at least one $S_{e,A}$ is not nowhere dense. Hence it is dense in some nonempty open set U . Choose $Y \in U$ and a positive rational r with $B_d(Y, r) \subseteq U$.

By (13), choose $C \leq_T Z$ with $\delta(C, X) < r/16$. Since $\delta(X, Y) = 0$, also $\delta(C, Y) < r/16$. Lemma 6.2 gives a finite modification $C_0 \leq_T Z$ such that

$$d(C_0, Y) < r/8. \quad (14)$$

Crucially, C_0 is *not* asserted to belong to $\mathcal{C}_X$.

To compute $A(n)$ from C_0 , search finite strings σ and convergent simulations $\Phi_e^\sigma(n) \downarrow = v$ subject to

$$\frac{|\{j < k : \sigma(j) \neq C_0(j)\}|}{k} < r/4 \quad (1 \leq k \leq |\sigma|). \quad (15)$$

This is an ordinary C_0 -computable search.

For termination, the dense set $S_{e,A}$ meets $B_d(Y, r/8) \subseteq U$. Choose D in that intersection. Then $d(D, C_0) \leq d(D, Y) + d(Y, C_0) < r/4$, so a finite prefix of D witnessing convergence on input n satisfies (15).

For correctness, take any accepted σ of length L and form $W = \sigma \star Y$, patching onto the genuine description Y , not onto C_0 . Then $W \in \mathcal{C}_X$. For $1 \leq k \leq L$, the prefix triangle inequality, (14), and (15) give

$$\rho_k(W \triangle Y) \leq \frac{|\{j < k : \sigma(j) \neq C_0(j)\}|}{k} + \rho_k(C_0 \triangle Y) < r/4 + r/8 = 3r/8.$$

After L there are no additional errors, so $d(W, Y) < 3r/8 < r$, by the finite-prefix identity. Thus $W \in U \cap [\sigma]$. This nonempty open set meets $S_{e,A}$, forcing the accepted value v to be $A(n)$.

We have computed A from $C_0 \leq_T Z$. Since A was any member of the core, $\mathcal{I}_X \subseteq \downarrow_T Z$. For computable Z , the opposite inclusion of all computable sets follows from Lemma 2.3. $\square$

Remark 6.4 (Why the external-center argument matters). A computable set can approximate X to every fixed positive error tolerance without any computable set being a coarse description of X . It would be incorrect to insert such an approximate set directly into a proof that assumes membership in $\mathcal{C}_X$. The two different patches in the proof above separate what the oracle computation can use from what the topological correctness argument can use.

7 An explicit complete c.e. counterexample

We now construct X rather than simply citing the existence of a random or generic example. Fix an acceptable enumeration (φ_e) of partial computable functions $\mathbb{N} \rightarrow \mathbb{N}$. It contains all total binary-valued computable functions. Let

$$R_e = \{n : \nu_2(n+1) = e\} = \{2^e(2k+1) - 1 : k \in \mathbb{N}\}, \quad (16)$$

where ν_2 is the exponent of 2 dividing a positive integer. These sets partition $\mathbb{N}$. Their densities and exact tail counts are

$$\rho(R_e) = 2^{-e-1}, \quad \bigcup_{e \geq k} R_e = \{n : 2^k \mid n+1\}, \quad \left| \left(\bigcup_{e \geq k} R_e \right) \cap [0, n] \right| = \left\lfloor \frac{n}{2^k} \right\rfloor. \quad (17)$$

Definition 7.1 (Guarded diagonal set). For $n \in R_e$, put

$$n \in X \iff [\forall m \leq n \varphi_e(m) \downarrow \in \{0, 1\}] \wedge \varphi_e(n) = 0. \quad (18)$$

The finite-prefix convergence guard is the point of the construction. Without it, a partial program might leave a noncomputable pattern inside one positive-density column. With the guard, every non-total or nonbinary program contributes only finitely many points to its column.

Theorem 7.2 (Properties of the guarded diagonal set). *The set X is computably enumerable and many-one complete. It has $\text{AP}_\emptyset(X)$, but it has no computable coarse description. Hence*

$$X \equiv_{\text{T}} \emptyset', \quad \mathcal{I}_X = \text{Comp}, \quad X \text{ is not coarsely computable.}$$

Proof. Computable enumerability. At stage s , examine each $n \leq s$, compute its unique column index e , and simulate the finitely many computations $\varphi_e(0), \dots, \varphi_e(n)$ for s steps. Enumerate n if all have halted with binary values and the last has value zero. An enumerated witness remains valid forever. Conversely, every witness to (18) is discovered at a finite stage. This is a c.e. enumeration of X .

Individual columns are computable. Fix e . If φ_e is total and binary-valued, then on R_e ,

$$X(n) = 1 - \varphi_e(n).$$

In this case $X \cap R_e$ is computable. Otherwise let m_e be the least input on which φ_e is undefined or has a nonbinary value. For every $n \geq m_e$, the guard in (18) fails, so $X \cap R_e \subseteq [0, m_e)$. This column is finite and hence computable. These alternatives prove computability for each fixed column, without an effective procedure deciding which alternative holds.

Arbitrarily accurate computable approximations. For fixed k , let

$$C_k = X \cap \bigcup_{e < k} R_e.$$

It is a finite union of computable sets, so it is computable. All errors of C_k lie in the tail union $\bigcup_{e \geq k} R_e$. Thus

$$\delta(C_k, X) \leq 2^{-k}. \quad (19)$$

For any positive ε , choose k with $2^{-k} < \varepsilon$. This proves $\text{AP}_\emptyset(X)$. It does not assert that indices for the C_k can be obtained effectively from k .

No computable coarse description. If C is a computable set, its characteristic function is φ_e for some total binary index e . Then $X(n) \neq C(n)$ for every $n \in R_e$. Therefore

$$\liminf_{n \rightarrow \infty} \rho_n(X \triangle C) \geq \rho(R_e) = 2^{-e-1} > 0.$$

So C is not a coarse description. A natural-number-valued computable coarse description would yield a binary one by the normalization in Section 2, and is excluded as well.

Many-one completeness. Let $K = \{p : \varphi_p(p) \downarrow\}$ be the usual halting set. By the parameter theorem, obtain computably from p an index $e(p)$ of the following program: on any input, first wait for $\varphi_p(p)$ to halt, and then return zero. If $p \in K$, this program is total and constantly zero. If $p \notin K$, it is nowhere defined. Put $n(p) = 2^{e(p)} - 1$, which belongs to $R_{e(p)}$. Then

$$p \in K \iff n(p) \in X.$$

The map $p \mapsto n(p)$ is total computable, giving $K \leq_m X$. As X is c.e., $X \leq_{\text{T}} K$, so $X \equiv_{\text{T}} \emptyset'$.

Finally, Theorem 6.3 applied to the proved approximability gives $\mathcal{I}_X = \text{Comp}$. $\square$

Remark 7.3 (An index-level subtlety). The displayed formula for C_k does not give a uniform algorithm for its characteristic function from k . The proof of each column's computability uses either a genuinely total program or the existence of a fixed finite set. Recognizing the appropriate case uniformly would decide information about program totality. No such recognition is used in this construction or in Theorem 6.3.

8 Minimal pairs, no least degree, and uncountable coinitality

From now on X denotes the explicit set of Definition 7.1.

Lemma 8.1 (The coarse classes are nonzero). *Neither $[X]_{\text{nc}}$ nor $[X]_{\text{uc}}$ contains a computable representative, including a computable natural-number-valued function.*

Proof. If a computable g belonged to either class, then $X \leq_{\text{nc}} g$. Since g is its own computable coarse description, the reduction would give a computable coarse description of X , contrary to Theorem 7.2. $\square$

Theorem 8.2 (The requested minimal-pair certificate). *There is $D \equiv_{\text{uc}} X$ such that $\deg_{\text{T}} D$ and $\mathbf{0}'$ form a Turing minimal pair. In particular both $[X]_{\text{nc}}$ and $[X]_{\text{uc}}$ have no least Turing degree.*

Proof. Theorem 4.5 gives $D \in \mathcal{C}_X$ with

$$(\downarrow_{\text{T}} D) \cap (\downarrow_{\text{T}} X) = \mathcal{I}_X = \text{Comp}.$$

Lemma 2.1 gives $D \equiv_{\text{uc}} X$, and Lemma 8.1 makes D noncomputable. The degree of X is $\mathbf{0}'$. Thus the two degrees are nonzero and their only common lower degree is $\mathbf{0}$, exactly the minimal-pair condition.

If g were a least representative of either coarse class, it would be computable from both X and D . Apply the common-lower-cone equality to the graph of g when g is a function. Then g would be computable, contradicting Lemma 8.1. $\square$

Corollary 8.3 (Perfect strengthening). *There is a perfect set $P \subseteq \mathcal{C}_X$ whose members form Turing minimal pairs with each other and, individually, with $\mathbf{0}'$. Every member is uniformly coarsely equivalent to X . In particular, every member is Turing incomparable with $\mathbf{0}'$ and is not Δ_2^0 .*

Proof. Apply Theorem 5.2 with prescribed oracle $B_0 = X$. The core is Comp , and no member of $\mathcal{C}_X$ is computable. The exact-pair equalities therefore become minimal-pair equalities. A nonzero degree forming a minimal pair with $\mathbf{0}'$ can neither be below nor above $\mathbf{0}'$; in particular no representative is computable from $\mathbf{0}'$. $\square$

Definition 8.4. For $r \in \{\text{nc}, \text{uc}\}$, write

$$\text{Spec}_{\text{T}}([X]_r) = \{\deg_{\text{T}} g : g \equiv_r X\}.$$

A subset Q of this spectrum is *coinital* if every degree in the spectrum bounds some degree in Q . This is the order-theoretic notion of coinitality, not a claim about a computable listing.

Theorem 8.5 (No countable coinital family). *For either $r \in \{\text{nc}, \text{uc}\}$ and every countable sequence $g_j \equiv_r X$, there is $D \in \mathcal{C}_X$ forming a Turing minimal pair with every g_j . Consequently $\text{Spec}_{\text{T}}([X]_r)$ has no countable coinital subset. Indeed one may choose a perfect family of such D , pairwise forming minimal pairs with each other.*

Proof. Let B_j be a set coding the graph of g_j , so that $B_j \equiv_{\text{T}} g_j$. Theorem 4.5, or its perfect strengthening, gives $D \in \mathcal{C}_X$ with

$$(\downarrow_{\text{T}} D) \cap (\downarrow_{\text{T}} B_j) = \mathcal{I}_X \cap (\downarrow_{\text{T}} B_j) = \text{Comp} \quad \text{for every } j.$$

All g_j and D are noncomputable by Lemma 8.1. Therefore the equalities are minimal-pair equalities. In particular $g_j \not\leq_{\text{T}} D$ for every j . Since $D \equiv_r X$, no degree in the proposed countable family lies below this member of the spectrum. The family is not coinital. $\square$

Remark 8.6 (Why this is stronger than the original target). A least degree would give a one-element cointial family. The theorem rules out even a countably infinite one, and does so with an oracle Turing incomparable with each proposed member. It does not exclude an uncountable set of minimal degrees. No exact cardinal value for the cointiality beyond uncountability is asserted here.

8.1 The representative spectrum is upward closed

For completeness, the familiar sparse-coding observation in the source report can be checked directly. Suppose $Y \equiv_r X$ is binary and $Y \leq_T B$. Define W by putting $W(2^n) = B(n)$ and keeping $W(k) = Y(k)$ at all other positions. The powers of two have density zero, so $W \equiv_{uc} Y$. Also $B \leq_T W$ by reading those positions, while $W \leq_T B$ because $Y \leq_T B$. Thus $\deg_T W = \deg_T B$.

The same argument works for function representatives by first choosing a set B of the desired higher degree and storing its bits on the powers of two. This shows that both full representative spectra are upward closed. There is no conflict with the preceding theorems: an upward-closed family of Turing degrees need not have a least member or a countable cointial basis.

9 Which nonuniform coarse classes have least representatives?

The least-degree property has a short structural characterization. We give it to clarify exactly why the earlier nonsurjectivity results already bear on C1. The proof uses a simple robust coding, with no uniform decoder claimed.

For $A \subseteq \mathbb{N}$, let $J(A)(0) = 0$ and define

$$J(A)(k) = A(n) \quad \text{when } 2^n \leq k < 2^{n+1}. \quad (20)$$

Thus bit $A(n)$ is repeated on a block of length 2^n .

Lemma 9.1 (Nonuniform robust recovery). *$J(A) \equiv_T A$, and every coarse description of $J(A)$ computes A . Consequently*

$$\mathcal{I}_{J(A)} = \downarrow_T A.$$

Proof. Computing $J(A)$ from A uses the block index. Conversely $A(n) = J(A)(2^n)$, proving Turing equivalence.

Let D be a binary coarse description of $J(A)$. Compute $b(n)$ as the majority value of D on $[2^n, 2^{n+1})$, resolving ties by zero. For all large n ,

$$|(D \triangle J(A)) \cap [0, 2^{n+1})| < \frac{1}{4} 2^{n+1} = 2^{n-1}.$$

This is fewer than half the positions in the n th block, so $b(n) = A(n)$ for all sufficiently large n . Thus A is a finite modification of the D -computable set b , and $A \leq_T D$. The finite correction depends on D ; there is no asserted uniform recovery of its cutoff or values.

All sets below A are therefore in the core. The reverse inclusion follows from $J(A) \in \mathcal{C}_{J(A)}$ and $J(A) \equiv_T A$. $\square$

Theorem 9.2 (Least-representative characterization). *A nonuniform coarse-equivalence class of total functions has a least Turing degree if and only if it is $[J(A)]_{nc}$ for some set A . In that case its least degree is $\deg_T A$.*

Proof. First suppose $g \equiv_{\text{nc}} J(A)$. Since $J(A) \leq_{\text{nc}} g$, the oracle g computes a coarse description D of $J(A)$. Lemma 9.1 yields $A \leq_{\text{T}} D \leq_{\text{T}} g$. The representative $J(A)$ itself has degree $\deg_{\text{T}} A$, so that degree is least in the class.

Conversely suppose the class has a least representative g , and choose a set $A \equiv_{\text{T}} g$ by graph coding. Every coarse description h of g is itself uniformly coarsely equivalent to g , by the disagreement triangle inequality. Leastness therefore gives $g \leq_{\text{T}} h$, and hence $A \leq_{\text{T}} h$. Thus every coarse description of g computes $J(A)$ itself, proving $J(A) \leq_{\text{nc}} g$.

In the other direction every coarse description of $J(A)$ computes A by Lemma 9.1, and hence computes g itself. Therefore $g \leq_{\text{nc}} J(A)$. This proves equality of the classes and identifies the least degree. $\square$

The map $\deg_{\text{T}} A \mapsto [J(A)]_{\text{nc}}$ is an order embedding: if $A \leq_{\text{T}} B$, every description of $J(B)$ computes B and then $J(A)$; if $J(A) \leq_{\text{nc}} J(B)$, apply the reduction to $J(B)$ itself to get $A \leq_{\text{T}} J(B) \equiv_{\text{T}} B$. Thus the characterization identifies exactly its image.

Remark 9.3 (Uniformity boundary). The theorem characterizes *nonuniform* coarse classes. The eventual majority decoding above does not provide one exact decoder for A that works on every description, and the converse proof does not establish a uniform coarse equivalence. No corresponding image characterization for uniform coarse classes is asserted. The negative uniform-class results in Section 8 do not depend on such a characterization.

10 Relativized examples with any principal core

The construction is not restricted to common information of computable degree. Fix any set Z . Using an acceptable enumeration of partial Z -computable functions, define X_Z by the guarded formula (18), with φ_e^Z in place of φ_e . The proof of Theorem 7.2 relativizes line by line:

$$X_Z \text{ is c.e. in } Z, \quad X_Z \equiv_{\text{T}} Z', \quad \text{AP}_Z(X_Z), \quad (21)$$

and X_Z has no Z -computable coarse description. The reductions proving $X_Z \equiv_{\text{T}} Z'$ have the same form: Z' enumerates membership in any Z -c.e. set, and the parameterized all-or-nothing programs reduce Z' to X_Z .

Let

$$Y_Z = J(Z) \oplus X_Z, \quad Y_Z(2n) = J(Z)(n), \quad Y_Z(2n+1) = X_Z(n). \quad (22)$$

Theorem 10.1 (Arbitrary principal cores). *For every Z the set Y_Z has*

$$Y_Z \equiv_{\text{T}} Z', \quad \mathcal{I}_{Y_Z} = \downarrow_{\text{T}} Z, \quad Y_Z \text{ has no } Z\text{-computable coarse description.}$$

Every representative of $[Y_Z]_{\text{nc}}$ or $[Y_Z]_{\text{uc}}$ has Turing degree strictly above $\deg_{\text{T}} Z$. Both spectra have no countable coinital family. There is a perfect family of representatives whose distinct pairs have greatest common lower degree exactly $\deg_{\text{T}} Z$.

Proof. Taking the odd half of Y_Z computes X_Z , and both halves are computable from Z' , proving $Y_Z \equiv_{\text{T}} Z'$. If D is a coarse description of Y_Z , its even half is a coarse description of $J(Z)$: up to n positions, its errors are bounded by the total errors of D up to $2n$, so their proportion is at most twice that total proportion. Lemma 9.1 gives $Z \leq_{\text{T}} D$. Thus $\downarrow_{\text{T}} Z \subseteq \mathcal{I}_{Y_Z}$.

Given $\varepsilon > 0$, choose a Z -computable C with $\delta(C, X_Z) < \varepsilon$. The join $J(Z) \oplus C$ is Z -computable, and only its odd half can disagree with Y_Z . Its upper discrepancy is $\frac{1}{2}\delta(C, X_Z) < \varepsilon/2$. Therefore $\text{AP}_Z(Y_Z)$ holds. Theorem 6.3 gives the opposite inclusion of cores, proving $\mathcal{I}_{Y_Z} = \downarrow_{\text{T}} Z$.

A Z -computable coarse description of Y_Z would have a Z -computable odd half describing X_Z , which is impossible. If g belongs to either coarse class, $Y_Z \leq_{\text{nc}} g$, so g computes a coarse

description of Y_Z , and hence computes Z . If also $g \leq_T Z$, that same description would be Z -computable, a contradiction. Thus every such g is strictly above Z .

For any countable family of representatives g_j , code them by oracles B_j . Theorem 4.5 gives a description D whose common lower cone with each B_j is $\downarrow_T Z$, since every B_j computes Z . Therefore $g_j \not\leq_T D$: otherwise g_j would be computable from Z , contrary to what was just proved. This excludes countable coinitality. Theorem 5.2 gives the perfect strengthening, with common lower cones exactly $\downarrow_T Z$. $\square$

This result concerns ordinary Turing degrees above $\deg_T Z$; it does not change the underlying definition of coarse reducibility. The parameter Z is used to construct an example whose common information is exactly that oracle's lower cone.

11 Proof audit, limitations, and continuation targets

11.1 Dependency ledger

The logical dependencies are short enough to expose explicitly:

prefix-density inequalities $\implies$ complete description space $\implies$ local recovery,
 local recovery + Baire category $\implies$ simultaneous exact partners,
 local recovery + finite-level fusion $\implies$ perfect exact pairs,
 approximability + Baire category + prefix search $\implies$ small core,
 guarded c.e. diagonalization + small core $\implies$ the explicit negative examples.

The many-one completeness statement additionally uses the standard parameter theorem for an acceptable enumeration. No randomness basis theorem, jump inversion theorem, classical priority minimal-pair construction, or uncountable effective listing is assumed.

11.2 Where incorrect shortcuts were excluded

The density metric δ cannot support the category proof: it vanishes throughout one description class. The supremum-prefix metric d is used instead, and its completeness and separability are proved.

Individual cone avoidance is not silently upgraded to simultaneous avoidance. The nowhere-density calculation precedes the countable Baire argument. The perfect construction explicitly handles all ordered pairs of distinct nodes at each level; separate pairwise existence assertions would not produce a single compatible perfect family.

The set used as a computable approximate center in Theorem 6.3 need not be a description. The algorithm queries that center, but the correctness proof patches onto a genuine description. This separation is indispensable.

The individual column programs in the c.e. construction are not presented as a uniformly computable sequence. The finite-correction argument in robust block decoding is also explicitly nonuniform. Neither step recovers unknown indices or stabilization stages effectively.

Finally, membership in the same coarse class is proved before applying the minimal-pair criterion, and nonzero coarse degree is proved independently. Ordinary noncomputability of the chosen binary set would not suffice.

11.3 Effectivity and formal verification

The set X is given by an explicit c.e. enumeration, with a proof that it is many-one complete. The additional descriptions and the perfect set are obtained by classical countable choices and Baire/fusion arguments. We do not provide a computable procedure for the perfect-set code, an oracle bound on that code, or a low arithmetical bound on its paths. In fact the selected paths forming minimal pairs with $\emptyset'$ cannot be Δ_2^0 .

No Lean executable was available in the working environment, and no Lean proof is supplied or claimed to compile. The source report is not treated as an axiom library. The mathematical proofs in this note are conventional proofs; the program in the archive checks only explicitly stated finite lemmas. Those checks are not evidence that the infinite theorems have been kernel verified.

11.4 The next mathematical questions

A meaningful continuation would determine how effectively the perfect family can be selected from a presentation of X and its core. The proof exposes the relevant obstacles: deciding that information lies outside the core, locating open rectangles outside closures of bad computation relations, and making compatible shrinkings. None of these choices has been assigned an arithmetical bound here. This is a newly proposed refinement, not a claim that it is a verified literature-open problem.

The nonprincipal-core case also deserves investigation. The perfect theorem already realizes any core arising from a coarse-description class as a common lower-cone ideal of pairs inside that same class. What additional restrictions characterize the ideals that arise this way is not determined here.

The effective-dense analogue cannot be inferred. An effective-dense description must either answer correctly or explicitly omit an answer; it cannot silently make wrong assertions. Density-zero modifications preserve coarse descriptions but do not in general preserve effective-dense descriptions. Thus the argument leaves C2 unchanged.

A Finite checks and reproduction

The archive contains one standard-library Python program, `checks/check_finite_lemmas.py`. It uses exact integers and rational fractions, with no floating-point density comparisons. The run packaged with this note completed **302,579 finite checks**, all passing.

Finite statement or model	Cases
Metric axioms on binary strings of length at most five	37,449
Exact finite-prefix patch identity	43,435
External-center patch inequality	9,021
Positive-density column counts	53,261
Tail-column counts	53,261
Column truncation error bounds	12,282
Guarded diagonalization on finite partial tables	13,120
Monotonicity of finite delayed enumerations	27,884
Strict-majority block recovery	52,866
Total	302,579

The finite partial tables use a distinguished value to mean *permanently undefined within the*

model, not a claim that a real program has failed to halt. The tests therefore do not decide totality. Similarly, finite column counts support the displayed arithmetic identities, not the Turing computability of individual infinite columns.

To rerun the checks and compile the article from the archive root:

```
python checks/check_finite_lemmas.py --output checks/results.json
pdflatex -interaction=nonstopmode -halt-on-error coarse_degree_attack.tex
pdflatex -interaction=nonstopmode -halt-on-error coarse_degree_attack.tex
```

Python 3.9 or later and a standard TeX Live installation suffice. The provided `build.sh` also performs these steps in an isolated build directory. The shipped PDF was compiled locally and rendered for visual inspection. The source report is included unchanged under `source/` for provenance; it is not needed to compile this article.

References

- [1] *Turing Degrees: A Unified Research Report*. Supplied file `turing_degrees_unified.tex`, unified edition 18 September 2026. In particular entry C1 and its minimal-pair diagnostic. An unchanged copy is included in this archive under `source/`.
- [2] Peter M. Gerdes. *Comparing Notions of Dense Computability on ω^ω and 2^ω* . arXiv:2508.06925v1, 9 August 2025. Definitions 2.3–2.4 and Question 7. <https://arxiv.org/abs/2508.06925v1>.
- [3] Denis R. Hirschfeldt, Carl G. Jockusch, Jr., Rutger Kuyper, and Paul E. Schupp. *Coarse Reducibility and Algorithmic Randomness*. Journal of Symbolic Logic **81**(3) (2016), 1028–1046. Preprint arXiv:1505.01707. In particular Definition 3.1 and Theorem 4.3. <https://arxiv.org/abs/1505.01707>. Author’s publication record: <https://www.math.uchicago.edu/~drh/Papers/coarsereducibility.html>.